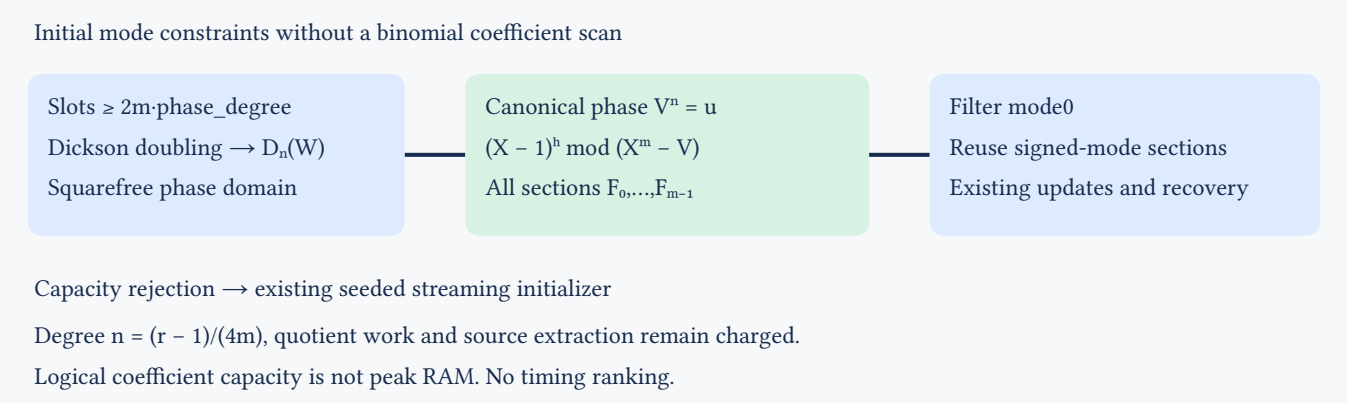
Initial mode constraints by quotient powering

T-1910 · Optional prime-field recovery initializer · 2026-11-08

Build the phase domain before mode0: construct $D_n(W)$ by Dickson doubling, take the squarefree phase support and impose $V^n=u$ canonically. Binary powering of $(X-1)^h$ modulo X^m-V then supplies all mode sections. Filter mode0 and reuse the signed sections in the smaller quotient. Successful powered initialization does not invoke the mode coefficient reader.



Phase capacity. For normalized character +1, the maximum squarefree W degree is floor(n/2)+1; for −1 it is n. The preflight checks twice m times that degree. Inversion pairs of V roots establish the degree formula, including fixed points V=±1. Rejected capacity uses the existing seeded streaming constructor. The slot bound covers logical section output, not peak memory.

Work retained. n=(r−1)/(4m). Dickson polynomials still reach degree n, and recursion depth is not total arithmetic complexity. Squarefree gcds, phase filtering, quotient products/reductions, allocations and scalar inverses remain. Sections are computed before mode0 can discard a branch. Source extraction, public-radius normalization, replay and point verification remain charged.

Representation constraint. Fixing the initial sign leaves L=(r−1)/(2m) m-th-power images. Excluding the possible endpoint leaves L−δ. Mode0 has at most m+1 sign-sum values, so some mode0-only fiber contains at least ceil((L−δ)/(m+1)) images. This bounds the degree of an explicit polynomial for that fiber, not typical retention or the full multi-mode recovery. It motivates combined constraints before large intermediate polynomials.

Sources. NIST DLMF18.5.1 and the predecessor reciprocal, canonical and powered-section modules. KB unavailable; exact-source fallback recorded. See DERIVATION.md for assumptions and the written proof. No new full Lean runtime refinement is claimed.

Qualification. 15,264 support comparisons and 2,529 public target/capacity cases pass with equal outcomes, routes and recorded source statistics. There are 132 Dickson equalities, 24 phase-degree checks and 12 exact/one-below capacity boundaries. 2,131 powered initializations and 4,160 fallbacks are coverage totals, not timings. Both revisions pass inherited regression. 120 copied sources and 134 dependencies are hash-bound. PCC-20261108-b7a0ed is unsigned producer evidence; independent review pending.

Provenance and next construction. The earlier 2mn-cap revision and changed sources are preserved. A shared-scope refusal was initially followed by an erroneous planning-log append; later shared writes waited for reservation. Next implement the derived half-degree squarefree factors for positive phases, avoiding their degree-n Dickson polynomial and squarefree gcd. These formulas are documented, not included as current savings.